



Sea-Bird Scientific
13431 NE 20th Street
Bellevue, WA 98005
USA

+1 425-643-9866
seabird@seabird.com
www.seabird.com

SENSOR SERIAL NUMBER: 1678
CALIBRATION DATE: 24-Feb-26

SBE 37 TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

COEFFICIENTS:

a0 = 2.763713e-05
a1 = 2.638070e-04
a2 = -1.517408e-06
a3 = 1.336822e-07

BATH TEMP (° C)	INSTRUMENT OUTPUT (counts)	INST TEMP (° C)	RESIDUAL (° C)
1.0000	744970.1	1.0000	0.0000
4.5000	637614.9	4.5000	-0.0000
15.0000	407528.0	14.9999	-0.0001
18.5000	353181.6	18.5001	0.0001
24.0000	283698.9	24.0003	0.0003
29.0000	233865.3	28.9995	-0.0005
32.5000	204942.0	32.5002	0.0002

n = Instrument Output (counts)

Temperature ITS-90 (°C) = $1 / \{a_0 + a_1[\ln(n)] + a_2[\ln^2(n)] + a_3[\ln^3(n)]\} - 273.15$

Residual (°C) = instrument temperature - bath temperature

